



PRODUCT NAME

Premium Organic Shower Gel Formula

Description

A cleansing, biodegradable shower gel with organic certified ingredients, designed to provide a pH balanced and environmentally responsible body wash experience.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	12 months
Batch Size:	50-100 liters
Processing Time:	Approximately 90 minutes
Temperature:	Room temperature (20°C - 25°C)
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, stirrer, pH meter
Certification:	Organic certified

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Aloe Vera Juice	Aloe Barbadensis Leaf Juice	30.0%	Moisturizing agent
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	12.0%	Mild surfactant
3	Decyl Glucoside	Decyl Glucoside	15.0%	Non-ionic surfactant
4	Glycerin	Glycerin	10.0%	Humectant
5	Xanthan Gum	Xanthan Gum	1.0%	Thickening agent
6	Citric Acid	Citric Acid	0.5%	pH adjuster
7	Potassium Sorbate	Potassium Sorbate	0.3%	Preservative
8	Sodium Benzoate	Sodium Benzoate	0.2%	Preservative
9	Organic Essential Oil Blend	Fragrance (Essential Oil)	1.0%	Fragrance
10	Purified Water	Aqua	30.0%	Diluent
Total			100%	

Manufacturing Process

Phase 1: Phase A - Aqueous Phase

1. Measure Purified Water into a mixing vessel.
2. Add Aloe Vera Juice and stir until well blended.

Phase 2: Phase B - Surfactant Blend

1. In a separate container, mix Cocamidopropyl Betaine and Decyl Glucoside.
2. Add Glycerin to the surfactant blend and incorporate completely.
3. Combine Phase A with the surfactant blend slowly while stirring continuously.

Phase 3: Phase C - Thickening and Hydration

1. Sprinkle Xanthan Gum into the mixture while stirring.
2. Allow the mixture to hydrate Xanthan Gum for 30 minutes.

Phase 4: Phase D - Final Adjustments

1. Add Citric Acid gradually and adjust to pH 7.
2. Incorporate Potassium Sorbate and Sodium Benzoate to preserve the formula.
3. Add Organic Essential Oil Blend for fragrance.

Required Equipment

Mixing vessel, stirrer, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Organic certified

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-100 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount to damp skin, then rinse thoroughly with water.